Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

ISO 14064 Internal Auditor Training

TGS Ref No: TGS-2025056357

Conducted by

Tertiary Infotech Academy Pte Ltd

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Version 3.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	Legacy release	Original trainer slide deck.	Tertiary Infotech Academy
2.0	19 March 2026	Prior Learner Guide and Lesson Plan release on the course Drive.	Tertiary Infotech Academy
3.0	20 August 2026	Complete source-aligned visual rebuild; expanded ISO 14064 audit, lifecycle, quantification, uncertainty and benchmarking coverage; ten connected activities and aligned WA/CS assessments.	Dr Alfred Ang

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Course Information

Course	ISO 14064 Internal Auditor Training
Course Code	TGS-2025056357
TSC	Carbon Footprint Management · PRE-ENV-5001-1.1
Duration	2 days · 16 training hours
Daily timing	9:00 am-6:00 pm; 60-minute lunch; two 10-minute breaks within training time
Assessment	Day 2, 4:00-6:00 pm · WA 1 hour + Case Study 1 hour

Learning Outcomes

- LO1: Develop process mappings and document GHG accounting reports aligned to ISO 14064 and international verification guidelines.
- LO2: Define lifecycle boundaries and determine emission sources to identify carbon reduction opportunities in line with assessment principles.
- LO3: Construct lifecycle inventory data for carbon footprint assessments using product configurations and scoped emission categories.
- LO4: Determine uncertainty factors and evaluate product carbon performance against industry benchmarks.

Course Schedule

Day 1 — ISO 14064 architecture, internal-audit practice, GHG principles, scopes and boundary decisions

Time	Duration	Topic / Activity	Slide reference	Method
9:00-9:30	30 min	Digital attendance, course orientation, introductions, learning outcomes and assessment briefing	Slides 1-19	Facilitated briefing and learner introduction
9:30-10:30	60 min	ISO 14064 family, ISO 14001 integration, verification principles and risk-based internal auditing	Slides 20-38	Interactive presentation and worked examples
10:30-10:40	10 min	Tea break	-	Break
10:40-11:25	45 min	Activity 1 — Map the GHG System and Audit Criteria	Slides 39-41	Process mapping and criteria crosswalk
11:25-12:10	45 min	Activity 2 — Plan a Risk-Based ISO 14064 Internal Audit	Slides 42-44	Audit planning and sampling workshop

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Time	Duration	Topic / Activity	Slide reference	Method
12:10-13:00	50 min	Activity 3 — Conduct Evidence Interviews and a Reporting Gap Review	Slides 45-48	Interview simulation, evidence testing and findings
13:00-14:00	60 min	Lunch break	-	Break
14:00-15:00	60 min	Emission scopes, gases, sources and sinks, organisational and lifecycle boundaries, hotspots and mitigation	Slides 49-67	Visual explanation and boundary cases
15:00-15:50	50 min	Activity 4 — Define Organisational and Product Lifecycle Boundaries	Slides 68-70	Boundary decision workshop
15:50-16:00	10 min	Tea break	-	Break
16:00-16:45	45 min	Activity 5 — Build the GHG Source, Sink and Scope Register	Slides 71-73	Scope mapping and significance review
16:45-17:30	45 min	Activity 6 — Diagnose Hotspots and Design a Mitigation Roadmap	Slides 74-76	Hotspot, sensitivity and roadmap workshop
17:30-18:00	30 min	Day 1 recap, Q&A and PM digital attendance	Slides 77-78	Retrieval practice and reflection

Total training time: 480 minutes (8 hours).

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Day 2 — Lifecycle inventory, quantification, uncertainty, benchmarking, reporting and audit close-out

Time	Duration	Topic / Activity	Slide reference	Method
9:00-9:15	15 min	Day 1 retrieval practice and AM digital attendance	Slide 78	Facilitated recap
9:15-10:15	60 min	Lifecycle phases, process mapping, LCI data, factors, calculation controls and software workflow	Slides 79-97	Visual explanation and worked examples
10:15-10:30	15 min	Activity 7 — Construct the Lumina 400 Lifecycle Inventory (part 1)	Slides 98-100	Unit-process and inventory setup
10:30-10:40	10 min	Tea break	-	Break
10:40-11:15	35 min	Activity 7 — Construct the Lumina 400 Lifecycle Inventory (part 2)	Slides 98-100	Data quality, balances and acceptance check
11:15-12:10	55 min	Activity 8 — Quantify and Audit the Product Carbon Footprint Model	Slides 101-104	Calculation, contribution and audit-trail workshop
12:10-13:00	50 min	Emission calculations, purchased energy, removals, uncertainty, benchmarking and audit findings (part 1)	Slides 105-123	Visual explanation and worked examples
13:00-14:00	60 min	Lunch break	-	Break
14:00-14:10	10 min	Emission calculations, uncertainty, benchmarking and audit findings (part 2)	Slides 105-123	Review and decision cases
14:10-14:55	45 min	Activity 9 — Evaluate Uncertainty and Product Benchmark Performance	Slides 124-126	Sensitivity and comparability workshop
14:55-15:20	25 min	Activity 10 — Close the Audit and Issue a Verification-Ready Report (part 1)	Slides 127-130	Finding calibration and closing-meeting simulation
15:20-15:30	10 min	Tea break	-	Break
15:30-15:45	15 min	Activity 10 — Close the Audit and Issue a Verification-Ready Report (part 2)	Slides 127-130	Report and follow-up acceptance check
15:45-16:00	15 min	Course wrap-up, LMS reminder and assessment briefing	Slides 131-138	Retrieval practice and transition to assessment

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Time	Duration	Topic / Activity	Slide reference	Method
16:00-17:00	**60 min**	**Written Assessment (WA-SAQ): 8 questions covering K1-K8**	**Slides 134-138**	**Individual, summative, open book**
17:00-18:00	**60 min**	**Case Study Assessment: 4 questions covering A1-A8; assessment attendance and submission**	**Slides 134-138**	**Individual, summative, open book**

Total training time: 480 minutes (8 hours).

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Activity and Competency Map

Activity	Title	Alignment	Evidence/output
1	Map the GHG System and Audit Criteria	K1 · K2 · K8 · A1 · A8 · LO1	GHG system process map; Criteria crosswalk; Document request list; Reporting and evidence index
2	Plan a Risk-Based ISO 14064 Internal Audit	K1 · K8 · A8 · LO1	Audit charter and scope; Risk assessment; Sampling plan; Interview timetable and opening agenda
3	Conduct Evidence Interviews and a Reporting Gap Review	K2 · K8 · A8 · LO1	Interview and corroboration records; Source-to-report audit trail; Recalculation samples; Finding register and issues briefing
4	Define Organisational and Product Lifecycle Boundaries	K3 · K5 · A1 · A2 · LO2	Organisational boundary diagram; Lifecycle process map; Functional unit and system boundary; Assumption and exclusion log
5	Build the GHG Source, Sink and Scope Register	K4 · K5 · A2 · A7 · LO2	Source, sink and reservoir register; Scope and ISO category map; Significance assessment; Data-owner and completeness matrix
6	Diagnose Hotspots and Design a Mitigation Roadmap	K2 · K3 · K4 · A7 · LO2	Hotspot and sensitivity dashboard; Mitigation option register; Prioritisation scorecard; 24-month roadmap and claim conditions
7	Construct the Lumina 400 Lifecycle Inventory	K3 · K4 · K5 · A1 · A3 · LO3	LCI data dictionary; Unit-process and flow register; Completed lifecycle inventory; Data-quality and gap-improvement plan
8	Quantify and Audit the Product Carbon Footprint Model	K2 · K4 · K5 · A3 · A4 · LO3	Controlled factor register; Quantified product footprint; Software/model setup record; Audit tests and correction bridge
9	Evaluate Uncertainty and Product Benchmark Performance	K6 · K7 · A5 · A6 · LO4	Uncertainty register; Sensitivity and scenario results; Benchmark comparability screen; Performance and claim recommendation
10	Close the Audit and Issue a Verification-Ready Report	K1 · K8 · A1 · A5 · A6 · A8 · LO1 · LO4	Final finding register; Closing presentation and meeting record; ISO 14064 internal-audit report; Corrective-action and follow-up plan

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